Paper code: CSE-501 Roll No...23BCS066.....

B. Tech. (Computer Engineering) V Semester Examination, 2025-26

Time: 3 Hours **Introduction to Data Mining** Max. Marks: 45

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Note: Attempt any two parts from each question.
Assume suitable missing data, if any.

Q1. (a) CO1 Describe in detail the architecture of data mining. Also discuss the different types of data on which data mining can be applied. [4.5] [BL2]

(b) Suppose your task as a software engineer at a Big-University is to design a data mining system to examine their university course database, which contains the following information: name, address and status (e.g.: undergraduate or graduate) of each student, the course taken, and their cumulative grade point average (GPA). Describe the architecture you choose. What is the purpose of each component of this architecture. [4.5] [BL3]

(c) Why is data preprocessing considered as an essential step in the data mining and machine learning process? Describe the major steps involved in data preprocessing. Explain the purpose and common techniques used in each step. [4.5] [BL2,3]

Q2. (a) CO2 (i) Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8), Compute the Euclidean distance and Manhattan distance between the two objects. [2+2.5= 4.5] [BL4,1]

(ii) Define the following terms:

a. Curse of Dimensionality
b. Underfitting and Overfitting

(b) Let the data matrix X be the score of students. Find two principal components from the following data matrix [4.5] [BL4]

Student	Math	DSA	Data Mining
1	90	60	90
2	90	90	30
3	60	60	60
4	60	60	90
5	30	30	30

(c) A data warehouse can be modeled by either a star schema or a snowflake schema. Briefly describe the similarities and the differences between the two models and then analyze their advantages and disadvantages regarding one another. [4.5] [BL2]

Q3. (a) Consider the following transactions. What association rules can be found in the set if the minimum support is 25% and the confidence is 60%. [4.5] [BL5]

CO3

Tid	Item-sets
T1.	F, A, D, B
T2.	D, A, C, E, B
T3.	C, A, B, E
T4.	B, A, D
T5.	D, B, C
T6.	A, B, D, E

(b) Consider the database shown in Table. Assuming the min_sup =3 enumerate all the frequent item-sets using the Eclat method. Show the output with the help of vertical database. [4.5] [BL5]

Table: Transaction DB

Tid	Item-sets
T1	A, B, C, D
T2	A, C, D, F
T3	A, C, D, E, G
T4	A, B, D, F
T5	B, C, G
T6	D, F, G
T7	A, B, G
T8	C, D, F, G

(c) How can we improve the performance of Apriori Algorithm. Using an example shows how Hash based techniques can be used in such a situation. [4.5] [BL2,3]

Q4. (a) Consider the following hypothetical data concerning student characteristics and whether each student should be hired. [4.5] [BL4]

CO4

Name	Attendance	Effort	GPA	Hirable
Sarah	Excellent	Lots	Average	Yes
Dana	Poor	Some	Average	No
Alex	Excellent	Some	Average	No
Annie	Poor	Lots	Poor	Yes
Emily	Excellent	Lots	Excellent	Yes
Pete	Poor	Lots	Excellent	No
John	Poor	Lots	Excellent	No
Kathy	Excellent	Some	Poor	No

Use Naive Bayes classifier to determine whether someone with Excellent attendance, poor GPA, and lots of effort should be hired. Discuss the solution of your answer.

e found in the set

Consider the following data, where the Y label is whether or not the child goes out to play [4.5] [BL3,4]

Day	Weather	Temperature	Humidity	Wind	Play?
1.	Sunny	Hot	High	Weak	No
2.	Cloudy	Hot	High	Weak	Yes
3.	Sunny	Mild	Normal	Strong	Yes
4.	Cloudy	Mild	High	Strong	Yes
5.	Rainy	Mild	High	Strong	No
6.	Rainy	Cool	Normal	Strong	No
7.	Rainy	Mild	High	Weak	Yes
8.	Sunny	Hot	High	Strong	No
9.	Cloudy	Hot	Normal	Weak	Yes
10.	Rainy	Mild	High	Strong	No
11.	Rainy	Cool	Normal	Strong	No
12.	Cloudy	Hot	High	Weak	Yes

Describe and calculate the information-gain for each attribute and choose which feature to split at Root in the Decision Tree.

(c) Explain linear SVM. Discuss advantages and disadvantages of SVM. [4.5] [BL2,3]

Q5. (a) CO5 Suppose that the data mining task is to cluster points (with(x,y) representing location) into three clusters, respectively. Where points are [4.5] [BL4]

A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9).

The distance function is Euclidean distance. Suppose initially we assign *A1*, *B1*, and *C1* as the centre of each cluster, respectively. Use the *k-means* algorithm to show the three clusters. Show all the rounds of execution for reaching the solution.

(b) (i) Why can DBSCAN produce clusters of arbitrary shape whereas partition-based algorithms fails to do that? [4.5] [BL3]

(ii) How does curse of dimensionality affect the distance-based clustering algorithms.

(c) Explain any one algorithm of hierarchical clustering. A dataset consisting of objects A, B, C, D, E with the following distance matrix is given: [4.5] [BL4]

Distance	A	B	C	D	E
A	0	9	2	3	1
B		0	8	6	5
C			0	7	10
D				0	4
E					0

Assuming single link hierarchical clustering is applied to the dataset? What dendogram will be returned?